

Chapter 7: The Reproductive Process

Nature of the Reproductive Process

- **Reproduction** is a fundamental property of life, enabling the continuity and evolution of animal populations.
- Two main modes: **asexual reproduction** and **sexual reproduction**.
- **Asexual reproduction** involves only one parent, producing genetically identical **clones**.
- **Sexual reproduction** involves two parents, each contributing **gametes** (sex cells), resulting in a genetically unique **zygote**.



- **Figure 7.1** illustrates types of asexual and sexual reproduction:
 - **Binary fission** in Paramecium produces two individuals.
 - **Budding** in Hydra forms new individuals from outgrowths.
 - **Hermaphroditism** in earthworms, where each individual has both male and female organs.
 - **Bisexual reproduction** in frogs, involving separate male and female individuals.
- **Asexual reproduction** is common in bacteria, unicellular eukaryotes, and many invertebrates (e.g., cnidarians, annelids, echinoderms).
- Main forms: **fission** (binary and multiple), **budding**, **gemmulation**, and **fragmentation**.
- **Binary fission**: Parent divides by mitosis into two equal parts.
- **Multiple fission**: Nucleus divides repeatedly before cytoplasm, producing many daughter cells.
- **Budding**: New individual arises as an outgrowth from the parent.
- **Gemmulation**: New individual forms from an aggregation of cells within a resistant capsule.
- **Fragmentation**: Animal breaks into parts, each capable of becoming a complete individual.

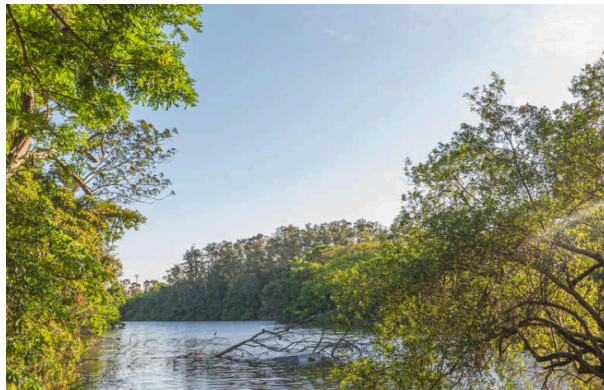
Sexual Reproduction

- **Sexual reproduction** produces individuals from **gametes**.
- Most common form is **bisexual reproduction**, involving two genetically different parents.
- **Dioecious** species have separate sexes; **monoecious** (hermaphroditic) species have both male and female organs in one individual.
- **Ovum** (egg): large, nonmotile, produced in small numbers by females.

- **Spermatozoon** (sperm): small, motile, produced in large numbers by males.
- **Meiosis** produces haploid gametes; **fertilization** restores diploid number.
- Genetic recombination during sexual reproduction increases variation and evolutionary potential.



- **Figure 7.2** shows the sexual life cycle: Haploid gametes combine to form a diploid zygote, which grows into an adult.
- **Hermaphroditism**: Both male and female organs in one individual; common in sessile or parasitic invertebrates.



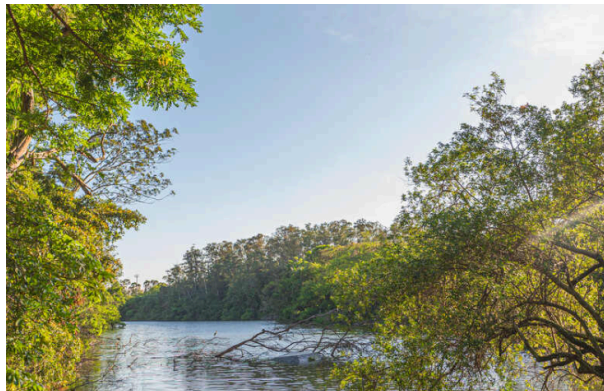
- **Figure 7.3** demonstrates hermaphroditic earthworms exchanging sperm.
- **Parthenogenesis**: Development of an embryo from an unfertilized egg; can be **ameiotic** (no meiosis, offspring are clones) or **meiotic** (haploid egg, may be activated by sperm or develop spontaneously).
- **Haplodiploidy**: In bees, fertilized eggs become diploid females, unfertilized eggs become haploid males.



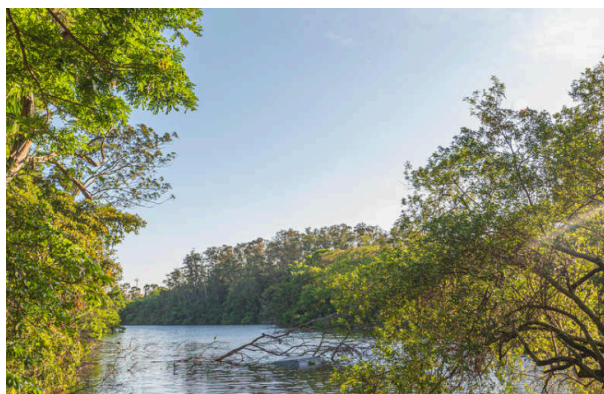
- **Figure 7.4** compares population growth in unisexual and bisexual lizards; unisexual populations grow faster as all individuals produce eggs.

Origin and Maturation of Germ Cells

- **Germ cells** form the **germ cell line**, providing continuity between generations.
- In vertebrates, **primordial germ cells** develop early and migrate to the gonads.



- **Figure 7.5** shows migration of primordial germ cells from the yolk sac to the genital ridges.
- **Sex determination** in mammals is chromosomal: SRY gene on the Y chromosome triggers testis formation and testosterone production.
- In some reptiles, sex is determined by incubation temperature.



- **Figure 7.6** illustrates temperature-dependent sex determination in reptiles.

Gametogenesis

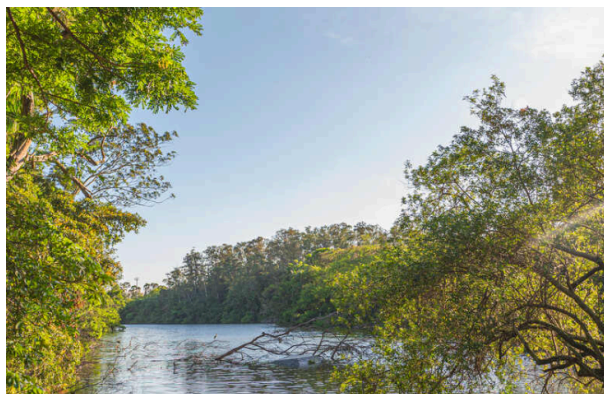
- **Gametogenesis** is the process of forming mature gametes.
- In males: **spermatogenesis**; in females: **oogenesis**.

Spermatogenesis

- Occurs in the **seminiferous tubules** of the testes.



- **Figure 7.7** shows a section of a seminiferous tubule with developing germ cells and mature sperm.
- **Spermatogonia** (diploid) divide by mitosis, become **primary spermatocytes**.
- Primary spermatocytes undergo meiosis I to form **secondary spermatocytes** (haploid).
- Secondary spermatocytes undergo meiosis II to form **spermatids** (haploid).
- Spermatids differentiate into **spermatozoa** (sperm).



- **Figure 7.8** details spermatogenesis, showing the stages from spermatogonia to mature sperm.



- **Figure 7.9** compares vertebrate and invertebrate sperm structure.

Oogenesis

- **Oogonia** (diploid) divide by mitosis, become **primary oocytes**.
- Primary oocytes begin meiosis but arrest in prophase I until puberty.
- Each menstrual cycle, one primary oocyte completes meiosis I, forming a large **secondary oocyte** and a small **first polar body**.
- Meiosis II produces an **ootid** and a **second polar body**; ootid matures into an **ovum**.
- Only one functional ovum is produced per primary oocyte.



- **Figure 7.10** illustrates oogenesis in humans, including meiotic divisions and polar body formation.

Reproductive Patterns

- **Oviparous:** Eggs laid outside the body; development occurs externally.
- **Ovoviviparous:** Eggs retained in the body; embryos nourished by yolk.
- **Viviparous:** Embryos develop within the mother, nourished directly by her.

Structure of Reproductive Systems

Invertebrate Reproductive Systems

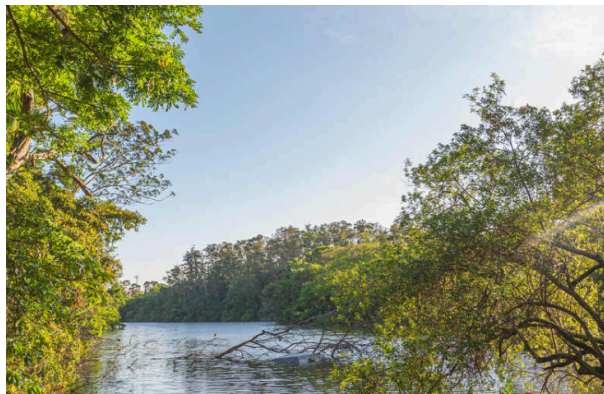
- Invertebrate systems vary from simple (gametes released into water) to complex (internal fertilization with specialized organs).



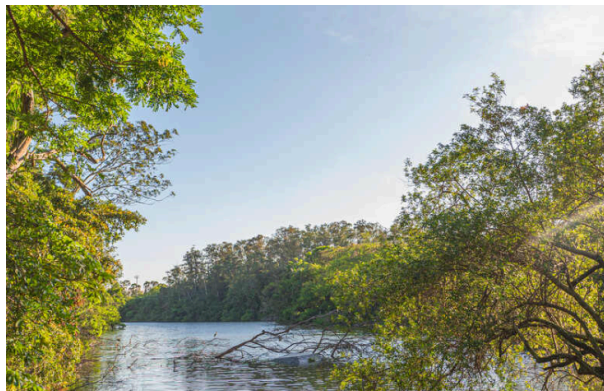
- **Figure 7.11** shows the reproductive system of crickets, including sperm transfer and storage.

Vertebrate Reproductive Systems

- Vertebrate reproductive and excretory systems are closely linked (urogenital system).
- Male system: **testes, seminiferous tubules, vasa efferentia, vas deferens, accessory glands, penis.**



- **Figure 7.12** shows the human male reproductive system.
- Female system: **ovaries, oviducts, uterus, vagina, external genitalia.**

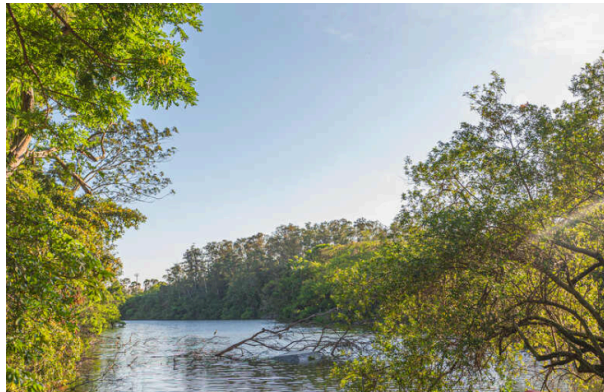


- **Figure 7.13** shows the human female reproductive system.

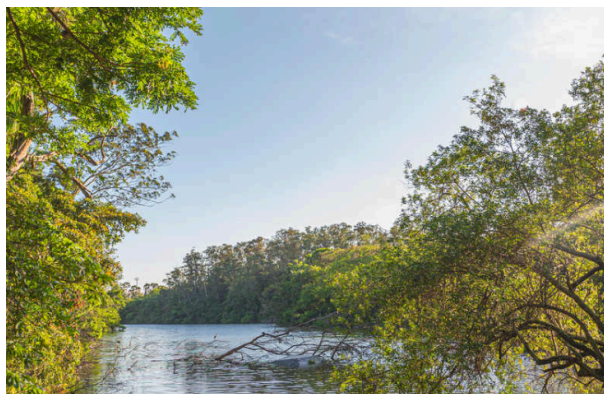
Endocrine Events in Reproduction

- Reproductive cycles are regulated by hormones, controlled by the **hypothalamus** and **anterior pituitary.**

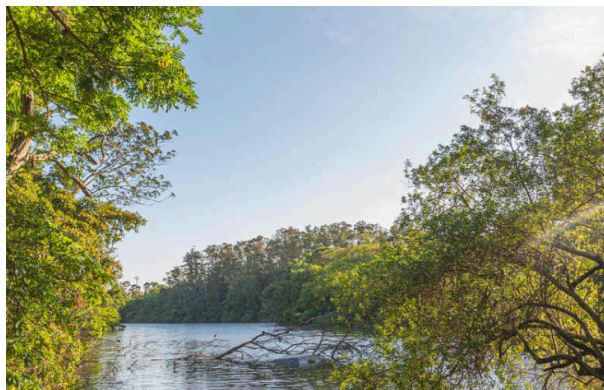
- Female cycles: **estrous** (most mammals) and **menstrual** (primates).
- Ovaries produce **estrogens** and **progesterone**; testes produce **testosterone** and **dihydrotestosterone (DHT)**.



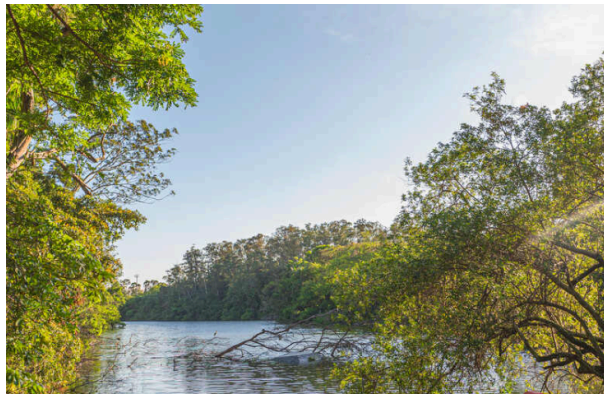
- **Figure 7.14** shows the chemical structures of sex hormones.
- Hormones regulate development of reproductive organs, secondary sex characteristics, and timing of reproduction.
- **GnRH** from the hypothalamus stimulates release of **FSH** and **LH** from the anterior pituitary.
- FSH and LH regulate gamete production and hormone secretion in the gonads.



- **Figure 7.15** illustrates the human menstrual cycle, including hormone levels and ovarian/uterine changes.
- During pregnancy, the **placenta** produces **hCG**, **estrogen**, **progesterone**, **hPL**, and other hormones.



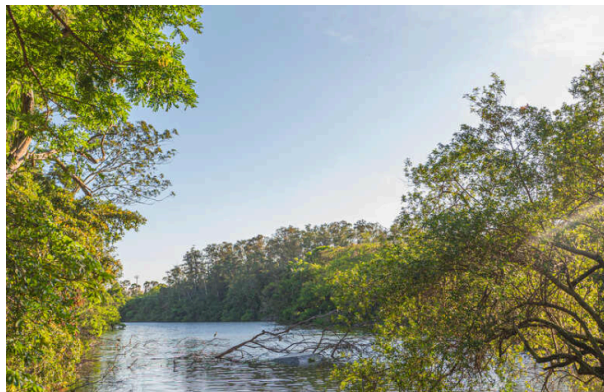
- **Figure 7.16** shows the roles of progesterone and estrogen in pregnancy.



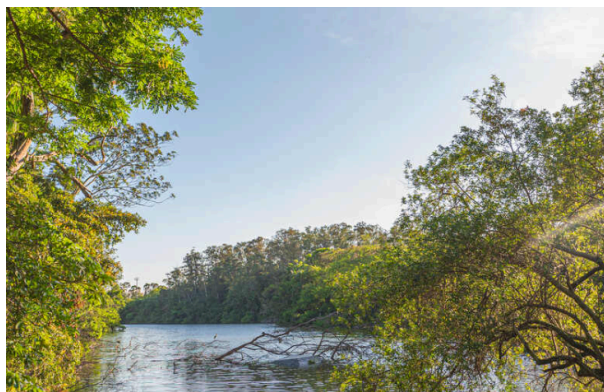
- **Figure 7.17** shows hormone levels during pregnancy.

Birth and Multiple Births

- **Birth** (parturition) involves dilation, expulsion, and placental delivery, regulated by hormones such as **oxytocin** and **prostaglandins**.



- **Figure 7.18** illustrates the stages of human birth.
- **Multiple births** can result from division of one zygote (identical twins) or from separate zygotes (fraternal twins).



- **Figure 7.19** shows the formation of human twins.

Key Terms and Concepts

- Asexual reproduction, sexual reproduction, gamete, zygote, clone, hermaphroditism, parthenogenesis, haplodiploidy, germ cell, gonad, meiosis, fertilization, spermatogenesis, oogenesis, oviparous, ovoviviparous, viviparous, estrous cycle, menstrual cycle, estrogen, progesterone, testosterone, GnRH, FSH, LH, hCG, placenta, oxytocin.

Important Formulas and Chemical Structures

- Chromosome number after meiosis:

$$n = \frac{N}{2}$$

- Example chemical formula: $C_{19}H_{28}O_2$ (testosterone)
- Example chemical formula: $C_{18}H_{24}O_2$ (estradiol)
- Example chemical formula: $C_{21}H_{30}O_2$ (progesterone)